Identify the correct wires

Purpose: prepare a reversible, passive CAN tap while retaining the original T-box. Print A4, actual size or fit to page. Diagrams are schematic, not photographs of your bike.

Do not connect by colour alone. The table below is a published QL-TBOX-JM reference, not a verified pinout for your particular Ultra Bee.

1. Record the bike and T-box

Bike year / version: _____

T-box model / revision: _____

Look for **QL-TBOX-JM** on the label. A different model, connector or wire arrangement needs its own documentation. Cover the IMEI, QR code and identifiers before sharing photos.

2. Locate the six-pin T-box connector

Before unplugging, follow the bike manufacturer's shutdown and battery isolation procedure. Key-off alone may leave the T-box powered. Photograph the connected harness first. Then record both sides of the connector with its latch visible.

Published pin	Function	Published wire colour
3	CAN-L	White with black stripe
6	CAN-H	White with red stripe
5	Ground / GND	Green with red stripe
2	Constant supply: 12 V	Yellow
4	Ignition / ACC: 12 V	Black with yellow stripe
1	Unassigned in table	Not specified

Source [1], section 2.2, page 4. The colour names describe the T-box pigtail in that manual; the bike loom side may differ. Supply and ACC are shown for identification only. Do not use them to power the ESP32 directly.

Find the tiny moulded cavity numbers. The wire-entry view and mating-face view are mirrored. Do not count left to right from an online photograph.

Photographs to take: label, connector mating face, wire-entry side, latch and cavity numbers. If the numbers cannot be read, leave the connections unassigned.

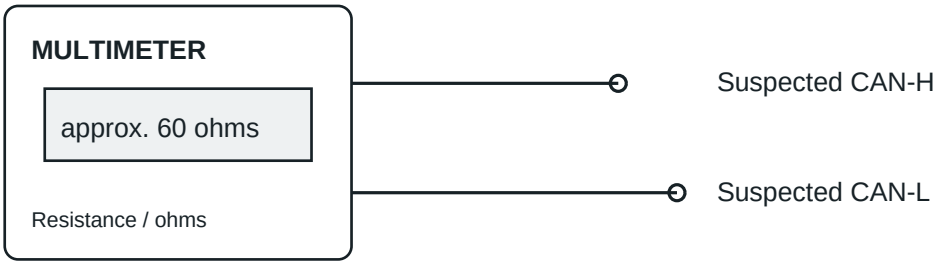
Confirm before connecting

Use insulated probes that fit without spreading terminals. Leave the logger disconnected during identification.

3. Check the suspected pair with power off

Keep away from traction terminals and motor phase wiring. With the charger disconnected, isolate the bike using its documented procedure. Check for zero voltage at the test points before selecting resistance mode. Never measure resistance on a powered circuit.

About 60 ohms across a fully connected, normally terminated CAN bus is supporting evidence only. It does not prove the pin identity, polarity or wiring variant. [3]



ALL POWER OFF - zero voltage confirmed first

If a connector is unplugged, it can remove a termination resistor and change the reading. A different value is a reason to investigate topology, not to add a resistor or guess another wire. Keep the OEM arrangement intact for the first capture.

4. Record the result

Check	Your observation
Actual CAN-H cavity / colour	_____
Actual CAN-L cavity / colour	_____
Verified signal ground / reference	_____
Zero voltage confirmed before ohms	_____
H-to-L resistance / connector state	_____

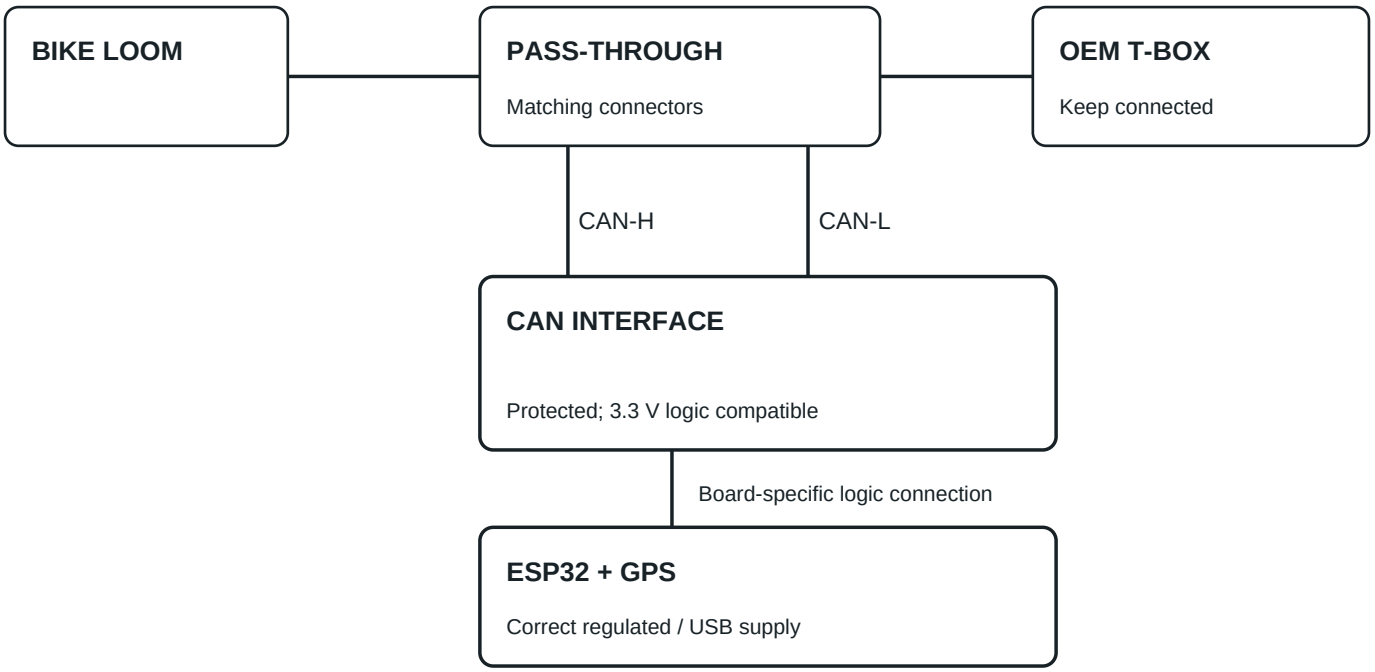
A multimeter alone cannot prove which line is H or L. If documentation and physical observations do not agree, have a vehicle electronics technician confirm the pair with suitable differential measurement equipment. Do not attach an earth-referenced oscilloscope ground clip to an unknown wire.

- ☐ Label and connector match the reference, with viewing direction recorded.
- ☐ Pin assignments confirmed from documentation and checks, not just colours.
- ☐ If anything is uncertain, stop here and obtain a confirmed pinout.

Fit a parallel CAN tap

Connection concept only. Exact ESP32 pins, transceiver supply, grounding and silent-mode wiring require the chosen board schematic.

CAN-H and CAN-L connect to a CAN transceiver/interface, never directly to ESP32 GPIO pins. An ESP32 development board alone is not a protected vehicle CAN interface. [2]



Ground/reference follows the interface design; not shown.

5. Prepare the interface on the bench

- ☐ Check supply, logic levels and free pins against the board schematic. Account for GPS and radio pin use.
- ☐ Configure true listen-only mode before attaching to the bike. It must suppress ACK and error transmissions, not merely avoid sending application messages. [2]
- ☐ If supported, set hardware silent mode per the datasheet. Verify passive behaviour at boot, reset and power loss.
- ☐ Disable the adapter's extra 120-ohm termination when joining the existing terminated bus. Do not remove factory termination. [3]

6. Assemble with the bike isolated

Use a reversible Y harness; retain the T-box. Keep the CAN branch short and twisted. Insulate and strain-relieve connections; protect from water, heat and moving parts.

Ground and power: for a non-isolated interface, connect its signal reference as specified to the verified bike signal ground. For an isolated interface, follow its bus-side ground and isolated-power requirements; do not bridge the isolation barrier. Do not assume the frame is signal ground. A USB power bank does not make a CAN interface isolated.

For initial tests, use a separate supply suitable for the logger. Do not connect T-box 12 V or traction-battery voltage directly to the ESP32. Permanent on-bike power needs a protected, fused regulator design.

First capture and work record

Stationary test only. No CAN requests, active diagnostics, T-box removal or test ride are needed for this first check.

7. Final checks before powering the bike

- ☐ All OEM connectors restored; no exposed conductors or accidental shorts.
- ☐ CAN-H, CAN-L and required reference checked against the recorded pinout.
- ☐ Logger supply and interface levels correct; extra termination disabled.
- ☐ Listen-only operation verified on the bench, including startup and reset.

8. Capture a short baseline

With the bike secured and throttle untouched, restore power using the bike's normal procedure. Check that the dashboard behaves normally. Log a short key-on trace with the original T-box connected. Record timestamps, CAN IDs, frame flags, payload bytes, capture settings and any error counters.

The bitrate is **not established for your bike in this guide**. Use a documented value or a passive/listen-only detection method. If no valid frames appear, isolate power before checking connections. Do not transmit probes or swap wires while live.

> **Stop if new dashboard faults appear, the bike behaves differently, a wire or module warms up, or communication errors increase. Power down, remove the added harness and restore the original arrangement before investigating.**

Logger / CAN interface model: _____

Firmware / listen-only setting: _____

Confirmed bitrate / source: _____

Capture filename / date / duration: _____

Notes / faults / follow-up:

Sources and scope

This guide supports identification and passive capture preparation. It is not a specimen-verified wiring drawing. No universal connector orientation, ESP32 pin allocation, CAN bitrate or decoded signal map is asserted.

[1] Surron QL-TBOX-JM user manual, section 2.2, page 4. Manufacturer document hosted by device.report.
https://device.report/m/0805101ae6f36a81e5dd0b39d12f594e8d1152e780c3ee6ed9882b4534274154

[2] Espressif ESP-IDF: TWAI hardware connection and listen-only mode.
https://docs.espressif.com/projects/esp-idf/en/stable/esp32/api-reference/peripherals/twai.html

[3] Kvaser: How to test if your CAN termination works correctly.
https://kvaser.com/developer-blog/how-to-test-your-can-termination-works-correctly/